

From Unequal Beginnings to Social Starting Capital

A Philosophical and Mathematical Framework for Intergenerational
Justice, Family Allocation, and Birthright Capital

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Abstract

This paper begins from a simple moral observation: human beings do not choose the circumstances of their birth, yet the distribution of starting conditions in the modern world is radically unequal. Some are born into wealth, assets, education, social capital, and legal security; others are born into precarity, debt, fragmented public services, and inherited disadvantage. The first part of the paper argues that this unevenness is neither a natural accident nor merely the outcome of individual effort. It is produced by history: enclosure, state formation, colonial extraction, slavery, inherited property, capital accumulation, and the conversion of common or public resources into exclusionary claims. Drawing on Locke, Rousseau, Paine, George, Marx, Polanyi, Rawls, Nozick, Sen, Ostrom, Van Parijs, Atkinson, and contemporary work on inequality and mobility, the paper develops a normative case for treating part of the social surplus of civilization as a common inheritance.

The second part introduces a formal proposal: the *Social Starting Capital Framework* (SSCF). The framework combines a universal child floor, a family allocation rule, a protected birthright capital account for each child, an education-linked staged unlock mechanism, and a sovereign social wealth fund financed by socially claimable flows such as resource rents, land value taxation, inheritance taxation, public asset profits, and ecological levies. The mathematical model is designed to satisfy three constraints at once: (i) every child must possess an irreducible claim on society's inherited wealth, (ii) family decisions should matter without annihilating the child's claim, and (iii) the system must remain demographically and fiscally sustainable. The paper provides definitions, formulas, examples, and basic propositions establishing non-zero child claims, bounded family expenditure growth, and a simple sustainability condition for the underlying fund. The result is a synthesis of political philosophy and formal policy design aimed at replacing the moral fiction of equal opportunity with an institutionally real starting stake for every person.

Part I

The Uneven World: History, Property, Capitalism, and the Moral Case for a Starting Stake

1 Introduction

One of the deepest asymmetries in social life is that the conditions into which persons are born are not chosen, yet they are decisive. A child may enter the world with access to secure housing, nutrition, healthcare, parental education, inheritance, family networks, and legal protection; another may enter with none of these. The first child is not necessarily more deserving; the second child is not necessarily less capable. The difference lies in the social structure into which each is inserted.

Much political language treats this situation as unfortunate but morally peripheral, as though justice requires only that later transactions be voluntary and formally legal. This paper rejects that view. If the initial distribution of life chances is already radically unequal, then a society that simply protects existing holdings and enforces contracts is not neutral; it is actively reproducing a historically inherited asymmetry. The problem is not merely poverty in the narrow sense. The deeper problem is that access to land, education, capital, time, mobility, and institutional power is pre-allocated before the individual begins to act.

Contemporary inequality data underscore the scale of the problem. According to the *World Inequality Report 2022*, the top 10% of the global population receives about 52% of global income, while the bottom 50% receives only 8.5%; in wealth, the top 10% owns roughly 76% while the bottom half owns around 2% (Chancel et al. [2022](#)). Intergenerational mobility is also highly uneven across countries, with poorer societies generally exhibiting weaker mobility and stronger dependence of outcomes on family background (World Bank [2021](#)). Land remains an especially concentrated asset class: estimates of global agricultural land inequality suggest that in some regions the top 10% of landowners control up to three quarters of agricultural land (Bauluz et al. [2020](#)). Meanwhile, land itself is finite and ecologically constrained; global land use remains heavily divided among agriculture, forests, built environments, and protected or conserved areas (Brivio and Tubiello [2025](#); United Nations Environment Programme [2024](#)).

The core question is therefore not whether people should be allowed to work, save, exchange, or create. They should. The question is whether a just society can allow radically unequal starting conditions to harden into a durable structure of inherited advantage. This paper

argues that it cannot. It then proposes a concrete framework through which every child receives a protected and socially funded starting stake, while preserving family agency, productive investment, and long-run fiscal sustainability.

2 The Birth Lottery and the Structure of Uneven Beginnings

There are many ways to describe inequality, but one of the clearest is the metaphor of the *birth lottery*. The moral force of the metaphor lies in its simplicity: no person chooses their parents, family wealth, nationality, neighborhood, school system, inherited health, or legal status at birth. These are morally arbitrary features in the Rawlsian sense (Rawls 1971). Yet these arbitrary features strongly shape later income, wealth, educational attainment, exposure to risk, and the ability to convert talent into durable advantage.

This point matters because modern defenses of inequality often move too quickly from the morality of *transactions* to the legitimacy of *distributions*. If one focuses only on whether later exchanges are voluntary, one can obscure the fact that the parties to these exchanges begin from radically different starting points. Equal formal liberty can coexist with vast material inequality if some people possess the time, capital, bargaining power, and fallback options that others lack.

The point is not that all inequality is unjust. Some inequalities may arise from choices, preferences, effort, luck, or socially valuable innovation. The point is that a social order cannot plausibly claim to be fair if the fundamental means of action are privately inherited by some and absent for others. The deeper injustice is therefore *pre-distributional*: the allocation of the basic conditions from which agency is exercised in the first place.

This is why the language of “equal opportunity” often rings hollow. Opportunity is not merely a legal permission. It is a function of capabilities, access, and security. A child with excellent formal rights but without nutrition, books, transport, healthcare, digital access, safe housing, or family stability does not stand in the same relation to opportunity as a child who possesses those things. Sen’s capability approach is helpful here: justice concerns what people are substantively able to be and do, not merely the formal rules under which they are addressed (Sen 1999).

A society serious about fairness must therefore ask a prior question: what material and institutional stake must each person possess simply in virtue of being born into a social world whose infrastructure, legal order, knowledge, and productive capacity were built across generations?

3 Historical Accumulation, Enclosure, and the Uneven World

The modern distribution of property did not arise from a blank slate. It emerged from a long history of enclosure, colonization, conquest, expropriation, inheritance, industrial concentration, and differential access to law and violence. The moral significance of that history is not that every current title is illegitimate in a simple or immediate sense. Rather, it is that present distributions are historically layered and path-dependent.

Locke famously grounded property in labor, arguing that individuals acquire rights by mixing their labor with nature, subject to a proviso that enough and as good be left for others (Locke 1689). That view continues to shape modern intuitions about effort and ownership. Yet even in Locke’s own framework, appropriation is not unconditional. Once land is scarce, enclosed, and legally defended, the relation between labor and ownership becomes less straightforward. The latecomer no longer confronts open nature but a world already partitioned.

Rousseau identified this moral fracture with remarkable clarity: the origin of civil inequality is not simply natural difference but the institution of property and the social recognition that follows from it (Rousseau 1755). Paine developed the point further in *Agrarian Justice*. He did not deny private cultivation or use, but argued that the earth in its natural state is the common property of the human race and that landed property therefore carries with it an obligation of compensation to those excluded from original use (Paine 1797). George later argued that the private capture of land rent allows socially generated value to be privately monopolized, even where the owner’s individual productive contribution is small (George 1879).

Marx and Polanyi extended the historical critique in distinct ways. Marx emphasized primitive accumulation, expropriation, and the structural power of capital (Marx 1867). Polanyi stressed that the modern market order required not the discovery of a naturally self-regulating system but a deliberate political project that transformed land, labor, and money into fictitious commodities (Polanyi 1944). On this account, the market is not merely a sphere of freedom. It is also a historically produced institution resting on prior acts of law, coercion, enclosure, and state formation.

Contemporary inequality research reinforces the historical point. Wealth is more enduring than income, and inherited wealth is especially powerful because it compounds across generations. Piketty’s work shows that capital ownership can become increasingly central to social stratification when returns to wealth outpace the growth of wages and output (Piketty 2014). The *World Inequality Report 2022* documents the continuing concentration of wealth at the global level (Chancel et al. 2022). Land inequality research similarly shows that unequal access to land remains a major driver of exclusion, particularly in agrarian and lower-income settings (Bauluz et al. 2020).

What follows is not that present holders are individually guilty for the entire past. It is instead that the moral baseline cannot be “whoever holds it now simply holds it rightfully because the law says so.” If social institutions rest on a long history of common inheritance, exclusion, and cumulative advantage, then some part of social wealth is not morally exhausted by existing private title.

4 Capitalism, Resources, and Competing Frameworks of Justice

4.1 Libertarian entitlement and its power

Nozick’s entitlement theory remains the most sophisticated modern defense of strong property rights and limited redistribution (Nozick 1974). In its most compelling form, the view does not require patterned outcomes; it requires just acquisition and just transfer. If holdings arise from legitimate initial acquisition and voluntary exchange, then the resulting distribution is just, even if highly unequal.

This framework captures an important truth: persons are not merely vessels for social planning, and the products of their labor, savings, and voluntary exchange deserve substantial protection. Any framework that treats all private wealth as simply available for confiscation is unstable both morally and economically. Incentives matter; security matters; family attachment matters.

Yet entitlement theory is most persuasive when the background structure is clean. In a world marked by inherited inequality, enclosed resources, legacy advantages, differential access to education and credit, and historically layered titles, the moral force of the theory weakens. Even Nozick acknowledged that historical injustice complicates rectification (Nozick 1974). The modern world is precisely such a complication.

4.2 Rawlsian justice and the arbitrariness of birth

Rawls offers a different baseline. Since natural endowments and social positions are morally arbitrary, institutions should be designed so that they would be chosen behind a veil of ignorance, where no one knows their place in society (Rawls 1971). This does not imply complete equality. It implies that inequalities are acceptable only if they are attached to positions open under fair equality of opportunity and are to the greatest benefit of the least advantaged.

Rawls is especially valuable because he shifts attention from market outcomes alone to the *basic structure* of society: the legal, educational, economic, and political institutions that shape life prospects before particular contracts are signed. The present paper is in that spirit. It asks not merely how to tax after the fact, but how to structure a fairer

starting point.

4.3 Capabilities, real freedom, and social inheritance

Sen’s capability approach insists that justice concerns effective freedom, not merely formal rights (Sen 1999). Van Parijs’s notion of *real freedom* pushes further: freedom requires resources sufficient to make options genuinely available, not merely legally permissible (Parijs 1995). A formally free but materially deprived person is often less free in any meaningful sense than a legally constrained but materially secure one.

This line of thought supports a starting-capital approach. The goal is not only to guarantee subsistence, but to furnish a person with the capacity to choose, refuse, relocate, study, invest, or build. Income support matters, but capital matters differently. Capital creates time horizons, resilience, bargaining power, and the ability to undertake projects rather than simply consume.

4.4 Commons, state capacity, and the design problem

Ostrom’s work reminds us that the choice is not simply between atomized private property and centralized command (Ostrom 1990). Common-pool resources can be governed by sophisticated institutions under appropriate conditions. This matters because the present proposal is not a call to abolish markets or family life. It is a call to recognize a *layer* of common inheritance and govern it institutionally.

That layer can be financed by resource rents, land value increments, inheritance taxation, public asset profits, and ecological charges. The OECD has documented the underused role inheritance taxation can play in addressing wealth concentration and improving efficiency (Organisation for Economic Co-operation and Development 2021). Existing sovereign and quasi-sovereign models also show that collective savings can be institutionalized. Norway’s Government Pension Fund was explicitly designed to transform petroleum wealth into intergenerational savings (Government of Norway 2025); Alaska’s Permanent Fund and dividend show that part of a resource flow can also be distributed directly to residents (State of Alaska, Permanent Fund Dividend Division 2026). These are not identical to the framework proposed here, but they demonstrate that intergenerational public capital is institutionally feasible.

4.5 From Paine and Atkinson to baby bonds

Paine proposed a national fund financed in part by ground-rent style reasoning to provide a lump sum at adulthood and an income in old age (Paine 1797). Atkinson revived a similar idea under the label of a minimum inheritance or capital endowment (Atkinson 2015). Hamilton and Darity’s baby bonds literature adds a modern asset-building perspective,

stressing how capital accounts can reduce deeply entrenched wealth gaps (Hamilton and William Darity 2010).

The present paper stands close to this tradition, but departs from it in one key respect. A purely per-child universal grant is normatively strong but fiscally exposed when demographic conditions shift. A purely per-family grant is fiscally stable but normatively weak because it allows the child's claim to be fully absorbed by parental fertility choices. The framework developed below is a hybrid: universal at the level of the child, family-sensitive at the margin, and anchored by a fund architecture rather than a one-time division of all existing wealth.

5 Why a New Framework is Needed

A compelling proposal must satisfy several conditions at once.

First, it must acknowledge that the planet, legal order, built infrastructure, social knowledge, and productive institutions are not created anew by each generation. Civilization is cumulative. There is therefore a plausible moral basis for a shared social inheritance.

Second, the framework must not collapse into the demand that all private wealth be liquidated and equally redistributed. That would be both politically implausible and economically destructive. The relevant target is not *all wealth*, but the *socially claimable surplus*: resource rents, land value increments, monopoly rents where appropriate, ecological charges, inheritance taxation, and public-asset income.

Third, the framework must protect the child as a moral unit. A child should not have zero starting claim simply because their parents have many children, poor planning, or low financial literacy.

Fourth, family agency must still matter. Households should retain real decision-making authority, and a system should not implicitly reward unlimited fertility under the assumption of infinite social resources.

Fifth, the system must preserve dynamic sustainability. A just starting stake cannot be funded by exhausting the capital base that would have supported future cohorts.

Sixth, education must be built into the design. Capital without capability can be dissipated under predation, desperation, or simple inexperience. But education should shape *how* funds become usable, not *whether* the basic claim exists.

These six conditions motivate the *Social Starting Capital Framework* (SSCF).

6 The Social Starting Capital Framework: Normative Statement

The SSCF rests on the following normative principles:

- P1. Common inheritance principle.** Part of social wealth is morally attributable not to current private effort alone but to nature, history, institutions, and cumulative social cooperation.
- P2. Irreducible child claim.** Every child has a non-zero and protected claim on that common inheritance.
- P3. Family agency principle.** Family decisions matter, but they may alter only the *marginal* distribution of support, not erase the child's basic claim.
- P4. Sustainability principle.** Current distributions must not destroy the capacity of future cohorts to receive an equivalent claim.
- P5. Capability principle.** Starting capital should be accompanied by education, staged access, and institutional protection.
- P6. Plural-institution principle.** Justice requires a bundle: services, income support where needed, capital access, and public asset stewardship.

The mathematical part of the paper translates these principles into a dynamic and budget-constrained formal system.

Part II

The Mathematical Framework: Definitions, Theory, and Examples

7 Preliminaries, Timing, and Core Notation

Time is discrete and indexed by years:

$$t = 0, 1, 2, \dots$$

Let:

- N_t = total population in year t ,
- $\mathcal{F}_t$ = set of eligible families (households) in year t ,
- $M_t = |\mathcal{F}_t|$ = number of eligible families in year t ,
- B_t = number of births in year t ,
- $K_{f,t}$ = number of dependent children in family $f \in \mathcal{F}_t$,
- $\mathcal{C}_{f,t}$ = set of dependent children in family f ,
- $\mathcal{B}_{f,t}$ = set of newborns in family f during year t .

To reduce volatility in grants caused by demographic spikes or short-run administrative noise, we define smoothed births:

$$\bar{B}_t = \frac{1}{q} \sum_{i=0}^{q-1} B_{t-i},$$

where $q \geq 1$ is the smoothing window (e.g. $q = 5$).

8 Socially Claimable Wealth and the Fund Architecture

The framework does *not* begin by dividing all private wealth by the population. That would be conceptually crude and economically destabilizing. Instead, it begins from the socially claimable component of wealth flows.

Table 1: Core symbols

Symbol	Meaning
N_t	Total population in year t
$\mathcal{F}_t$	Set of eligible families/households in year t
M_t	Number of eligible families
$B_t, \bar{B}_t$	Births in year t ; smoothed births
$K_{f,t}$	Dependent children in family f
Y_t	Socially claimable wealth inflow in year t
W_t	Value of the social wealth fund at start of year t
D_t^Y, D_t^W	Distribution financed from current inflows; distribution financed from fund draw
U_t	Universal child-floor expenditure
H_t	Total household allocation expenditure
Ψ_t	Birthright pool for newborn accounts
$g_{f,c,t}^{(0)}$	Initial grant assigned to newborn c in family f in year t
$A_{c,t}$	Protected account value of child c in year t

Definition 8.1 (Socially claimable inflow). *Let*

$$Y_t = R_t + L_t + I_t + P_t + C_t + X_t,$$

where:

$R_t = \text{resource rents},$

$L_t = \text{land value tax revenue},$

$I_t = \text{inheritance/estate/gift tax revenue dedicated to the framework},$

$P_t = \text{profits from public assets or public investment authorities},$

$C_t = \text{carbon, extraction, or ecological charges},$

$X_t = \text{other dedicated revenues}.$

A share $\phi \in [0, 1]$ of the inflow is saved into a social wealth fund, and the remainder may be used contemporaneously:

$$\phi Y_t \quad \text{saved,} \quad (1 - \phi) Y_t \quad \text{available for current distribution.}$$

Let W_t denote the value of the fund at the *start* of year t , and r_t its net return during the year.

We separate current-flow distribution from fund-based distribution:

$$D_t = D_t^Y + D_t^W,$$

with feasibility conditions

$$D_t^Y \leq (1 - \phi)Y_t, \quad D_t^W \leq \gamma W_t,$$

where $\gamma \in [0, 1]$ is the safe draw rate from the fund.

Hence the maximum total distributable amount in year t is

$$D_t^{\max} = (1 - \phi)Y_t + \gamma W_t.$$

The fund evolves according to

$$W_{t+1} = W_t(1 + r_t) + \phi Y_t - D_t^W.$$

Remark 8.1. *This separation is important. It prevents accounting confusion between what is spent directly from yearly inflows and what is withdrawn from the long-run capital base.*

9 Distribution Layers

The framework has four possible layers:

$$D_t = U_t + H_t + G_t + V_t,$$

where:

U_t = universal child floor,

H_t = household/family allocation,

G_t = birthright capital assigned to child accounts,

V_t = optional adult citizen dividend or transition support.

In the baseline version of the model, V_t may be set to zero:

$$V_t = 0.$$

The policy priority order is:

universal child floor $\rightarrow$ household allocation $\rightarrow$ birthright accounts $\rightarrow$ optional adult layer.

This ordering reflects the philosophical claim that subsistence and developmental support for children are more fundamental than discretionary adult distributions.

10 Universal Child Floor

Let u_t be the minimum annual support per dependent child. This need not be cash. It can be implemented through services, vouchers, or guaranteed provision:

- nutrition,
- healthcare,
- schooling,
- digital access,
- transport,
- housing support,
- basic developmental services.

The total annual universal child floor is

$$U_t = u_t \sum_{f \in \mathcal{F}_t} K_{f,t}.$$

This term is deliberately independent of household wealth and fertility strategy at the level of the child's minimum floor. It operationalizes the principle that no child may fall below the basic developmental threshold merely because of household circumstances.

11 Household Allocation: The Family Layer

A purely child-based system risks losing sight of the household as the actual site of care, expenditure, and planning. We therefore include a family allocation.

Each family receives a base amount h_t , plus a supplement tied to the number of dependent children via a weighted function:

$$H_{f,t} = h_t + s_t \omega(K_{f,t}),$$

where s_t is the supplement rate and $\omega(\cdot)$ is a non-linear child-weight function.

Definition 11.1 (Weighted child function). *Let*

$$\omega(K) = \sum_{j=1}^K w_j,$$

with weights satisfying

$$w_1 \geq w_2 \geq w_3 \geq \cdots \geq 0.$$

A useful and tractable specification is:

$$w_j = \begin{cases} 1, & j \leq K^*, \\ \lambda^{j-K^*}, & j > K^*, \end{cases} \quad 0 < \lambda < 1.$$

Interpretation:

- the first K^* children receive full household-weight support;
- subsequent children still count, but at a tapering rate;
- the family is neither ignored nor given an indefinitely linear claim on public resources.

The total household allocation is

$$H_t = \sum_{f \in \mathcal{F}_t} H_{f,t} = \sum_{f \in \mathcal{F}_t} (h_t + s_t \omega(K_{f,t})).$$

Proposition 11.1 (Bounded household supplement). *Suppose w_j is defined by the piecewise rule above. Then for every family,*

$$\omega(K) \leq K^* + \frac{\lambda}{1 - \lambda}.$$

Consequently,

$$H_{f,t} \leq h_t + s_t \left(K^* + \frac{\lambda}{1 - \lambda} \right).$$

Proof. For $K \leq K^*$, the claim is immediate because $\omega(K) = K \leq K^*$. For $K > K^*$,

$$\omega(K) = K^* + \sum_{m=1}^{K-K^*} \lambda^m \leq K^* + \sum_{m=1}^{\infty} \lambda^m = K^* + \frac{\lambda}{1 - \lambda}.$$

Multiplying by s_t and adding h_t yields the stated bound. $\square$

This proposition is one of the key technical reasons for adopting a tapered family rule rather than a simple linear-per-child household grant.

12 Residual Space and the Birthright Pool

Once the mandatory child floor and household layer are funded, the remaining distribution space in year t is

$$\mathcal{R}_t = D_t^{\max} - U_t - H_t - V_t.$$

The birthright pool is a share of this residual:

$$\Psi_t = \max\{0, \theta_G \mathcal{R}_t\}, \quad 0 \leq \theta_G \leq 1.$$

If $V_t = 0$, this simplifies to

$$\Psi_t = \max\{0, \theta_G (D_t^{\max} - U_t - H_t)\}.$$

The raw per-newborn birthright grant is then

$$g_t^{\text{raw}} = \frac{\Psi_t}{B_t}.$$

Remark 12.1. *The system therefore does not promise a fixed grant regardless of demographics and macro conditions. It promises a fair share of a sustainably distributable pool.*

13 Demographic Adjustment

Let the population growth rate be

$$n_t = \frac{N_{t+1} - N_t}{N_t},$$

and the realized fund growth rate be

$$v_t = \frac{W_{t+1} - W_t}{W_t}.$$

We define a demographic stabilizer:

$$\Delta_t = \min \left\{ 1, \frac{1 + v_t}{1 + n_t} \right\}.$$

The demographically adjusted base grant becomes

$$g_t^{\text{adj}} = \Delta_t g_t^{\text{raw}}.$$

This adjustment is conservative:

- if fund growth keeps pace with population growth, Δ_t stays close to 1;
- if population grows faster than the fund, new cohort grants are moderated rather than the system becoming insolvent.

14 The Hybrid Child Grant: Universal and Family-Sensitive Components

A purely per-family allocation would make the child's claim too dependent on parental fertility choices. A purely per-child allocation would ignore the household as a budgetary unit. We therefore split the newborn grant into two components.

Let $\alpha \in [0, 1]$ be the universal protection share. Define

$$g_{f,c,t}^{(0)} = \alpha g_t^{\text{adj}} + (1 - \alpha) g_t^{\text{adj}} \rho(K_{f,t}),$$

where $\rho(K)$ is a family-size adjustment factor. A simple choice is

$$\rho(K) = \frac{1}{1 + \mu(K - 1)}, \quad \mu \geq 0.$$

Interpretation:

- αg_t^{adj} is the child's irreducible universal claim;
- the second term preserves family sensitivity without collapsing the child claim to zero.

Proposition 14.1 (Lower bound on the child claim). *If $g_t^{\text{adj}} > 0$ and $\alpha > 0$, then for every newborn c in family f ,*

$$g_{f,c,t}^{(0)} \geq \alpha g_t^{\text{adj}}.$$

Proof. Since $\rho(K) \geq 0$ for all $K \geq 1$, the second term in

$$g_{f,c,t}^{(0)} = \alpha g_t^{\text{adj}} + (1 - \alpha) g_t^{\text{adj}} \rho(K_{f,t})$$

is nonnegative. Therefore

$$g_{f,c,t}^{(0)} \geq \alpha g_t^{\text{adj}}.$$

□

This proposition formalizes the moral principle that no child's starting stake is allowed to vanish simply because of family size.

15 Child Accounts and Accumulation

Each newborn c receives a protected account:

$$A_{c,t}^{(0)} = g_{f,c,t}^{(0)}.$$

The account grows with investment returns and may also receive yearly top-ups:

$$A_{c,t+1} = A_{c,t}(1 + q_t) + \tau_{f,c,t},$$

where:

q_t = net credited return on child accounts,

$\tau_{f,c,t}$ = annual top-up for child c .

A progressive top-up rule may be defined by household income:

$$\tau_{f,c,t} = \tau_t^{\text{base}} \left[1 + \kappa \max \left\{ 0, \frac{y^* - y_{f,t}}{y^*} \right\} \right],$$

where:

$y_{f,t}$ = household income of family f ,

y^* = reference income threshold,

$\kappa \geq 0$ is the progressivity parameter.

Thus:

- universality is preserved at the base;
- poorer families receive larger top-ups where desired;
- the account evolves as capital, not merely as consumption support.

If top-ups are omitted, then for constant q ,

$$A_{c,t+m} = g_{f,c,t}^{(0)}(1 + q)^m.$$

16 Education, Capability, and Staged Access

The framework does not treat financial education as a condition for deservingness. Instead, it treats education as a condition for flexibility.

Let $a_1 < a_2 < a_3$ be release ages, for example

$$a_1 = 18, \quad a_2 = 21, \quad a_3 = 25.$$

Let the release shares satisfy

$$\eta_1 + \eta_2 + \eta_3 = 1, \quad \eta_j \geq 0.$$

Then the nominal tranche available at age a_j is

$$U_c^{(j)} = \eta_j A_{c,t+a_j}.$$

Now define an education/capability score

$$E_{c,t} = \sum_{m=1}^J \beta_m z_{c,m,t},$$

where:

$$\begin{aligned} z_{c,m,t} &\in \{0, 1\} \quad \text{indicates completion of module } m, \\ \beta_m &\geq 0, \quad \sum_{m=1}^J \beta_m = 1. \end{aligned}$$

Modules may include:

- budgeting,
- savings and investment basics,
- debt and interest,
- fraud and predation awareness,
- contracts and housing,
- entrepreneurship,
- family planning and long-run cost awareness.

Let $\delta \in [0, 1]$ be the guaranteed discretionary fraction even without full completion. Then the immediately discretionary amount is

$$\tilde{U}_c^{(j)} = U_c^{(j)} \left[\delta + (1 - \delta) E_{c,t+a_j} \right].$$

The remainder,

$$U_c^{(j)} - \tilde{U}_c^{(j)},$$

is not forfeited. It stays in protected or restricted form (e.g. education, housing, tools, or approved entrepreneurial uses) until later release.

This avoids a punitive design while still embedding developmental structure.

17 System Feasibility and Sustainability

The annual budget constraint is

$$U_t + H_t + G_t + V_t \leq D_t^{\max},$$

or in the baseline version,

$$U_t + H_t + G_t \leq D_t^{\max}.$$

Since

$$G_t = \sum_{f \in \mathcal{F}_t} \sum_{c \in \mathcal{B}_{f,t}} g_{f,c,t}^{(0)} + \sum_{f \in \mathcal{F}_t} \sum_{c \in \mathcal{C}_{f,t}} \tau_{f,c,t},$$

the planner must ensure that all chosen parameters remain within this bound.

A simple sufficient condition for the long-run non-decline of the fund is the following.

Proposition 17.1 (Non-decline condition for the fund). *Suppose that in year t ,*

$$D_t^W \leq \gamma W_t \quad \text{and} \quad r_t \geq \gamma.$$

Then

$$W_{t+1} \geq W_t + \phi Y_t.$$

In particular, the fund does not shrink relative to its opening level when the return meets or exceeds the safe draw rate.

Proof. Using the law of motion,

$$W_{t+1} = W_t(1 + r_t) + \phi Y_t - D_t^W.$$

Since $D_t^W \leq \gamma W_t$,

$$W_{t+1} \geq W_t(1 + r_t - \gamma) + \phi Y_t.$$

If $r_t \geq \gamma$, then $1 + r_t - \gamma \geq 1$, so

$$W_{t+1} \geq W_t + \phi Y_t.$$

□

Remark 17.1. *This proposition is deliberately simple. Real policy would use expected returns, moving averages, prudential buffers, and stress testing rather than a one-period deterministic condition. But it captures the basic idea: the distribution rule should not consume principal faster than the fund can reasonably sustain.*

18 A Toy Calibration

This section provides illustrative calculations using arbitrary currency units (CU). The numbers are not intended as empirical estimates for any particular country; they are included to show the framework's behavior.

18.1 Baseline macro parameters

Assume:

$$\begin{aligned}Y_t &= 100 \text{ billion CU,} \\W_t &= 500 \text{ billion CU,} \\ \phi &= 0.40, \\ \gamma &= 0.03, \\ r_t &= 0.05.\end{aligned}$$

Then:

$$D_t^{\max} = (1 - \phi)Y_t + \gamma W_t = 0.60(100) + 0.03(500) = 60 + 15 = 75 \text{ billion CU.}$$

If the full safe draw is used, the next-period fund becomes

$$W_{t+1} = 500(1.05) + 40 - 15 = 550 \text{ billion CU.}$$

So even after a safe draw, the fund grows in this calibration.

18.2 Child floor and household layer

Assume:

$$\begin{aligned}U_t &= 25 \text{ billion CU,} \\H_t &= 20 \text{ billion CU,} \\V_t &= 0.\end{aligned}$$

Residual distributable space:

$$\mathcal{R}_t = 75 - 25 - 20 = 30 \text{ billion CU.}$$

Let $\theta_G = 0.50$. Then

$$\Psi_t = 0.50 \times 30 = 15 \text{ billion CU.}$$

If smoothed births are

$$\bar{B}_t = 500,000,$$

then the raw newborn grant is

$$g_t^{\text{raw}} = \frac{15,000,000,000}{500,000} = 30,000 \text{ CU.}$$

If $\Delta_t = 1$, then

$$g_t^{\text{adj}} = 30,000.$$

18.3 Family-size adjustment

Now assume

$$\alpha = 0.70, \quad \mu = 0.25.$$

Then for a family with $K = 1$,

$$\rho(1) = \frac{1}{1 + 0.25(1 - 1)} = 1,$$

so

$$g^{(0)} = 0.70(30,000) + 0.30(30,000)(1) = 30,000.$$

For a family with $K = 3$,

$$\rho(3) = \frac{1}{1 + 0.25(2)} = \frac{1}{1.5} \approx 0.6667,$$

hence

$$g^{(0)} \approx 21,000 + 9,000(0.6667) \approx 27,000.$$

For a family with $K = 6$,

$$\rho(6) = \frac{1}{1 + 0.25(5)} = \frac{1}{2.25} \approx 0.4444,$$

hence

$$g^{(0)} \approx 21,000 + 9,000(0.4444) \approx 25,000.$$

This captures the intended structure:

- larger families do not erase the child's basic claim;
- larger families do slightly dilute the family-sensitive component;
- the universal protected share remains intact.

18.4 Household support example

Assume:

$$h_t = 5,000, \quad s_t = 4,000, \quad K^* = 2, \quad \lambda = 0.60.$$

Then:

$$\omega(1) = 1, \quad \omega(3) = 1 + 1 + 0.6 = 2.6,$$

and

$$\omega(6) = 1 + 1 + 0.6 + 0.36 + 0.216 + 0.1296 = 3.3056.$$

Therefore:

$$H_{f,t}(K = 1) = 5,000 + 4,000(1) = 9,000,$$

$$H_{f,t}(K = 3) = 5,000 + 4,000(2.6) = 15,400,$$

$$H_{f,t}(K = 6) = 5,000 + 4,000(3.3056) = 18,222.4.$$

The family allocation rises with family size, but less than linearly. This is exactly what budgetary discipline requires.

18.5 Account growth to adulthood

Assume no annual top-ups and a constant net credited return $q = 0.04$. Then after 18 years:

$$A_{c,t+18} = g_{f,c,t}^{(0)}(1.04)^{18}.$$

Since

$$(1.04)^{18} \approx 2.0258,$$

we obtain:

$$30,000 \times (1.04)^{18} \approx 60,774.50,$$

$$27,000 \times (1.04)^{18} \approx 54,697.05,$$

$$25,000 \times (1.04)^{18} \approx 50,645.41.$$

Thus even with family-sensitive adjustment, the child retains a meaningful adulthood stake.

19 A Demographic Shock Example

Suppose that annual births increase, but the five-year smoothed value rises only from

$$\bar{B}_t = 500,000 \quad \text{to} \quad \bar{B}_{t+1} = 550,000.$$

Holding $\Psi_{t+1} = 15$ billion CU fixed for simplicity,

$$g_{t+1}^{\text{raw}} = \frac{15,000,000,000}{550,000} \approx 27,272.73.$$

If the fund also grows and $\Delta_{t+1} \approx 0.98$, then

$$g_{t+1}^{\text{adj}} \approx 26,727.27.$$

Compared with the original 30,000, the new cohort grant declines, but smoothly rather than catastrophically. This is the function of smoothing and demographic adjustment: they convert demographic pressure into gradual parameter movement instead of systemic instability.

20 Comparative Statics and Interpretation

The framework contains a number of interpretable parameters.

20.1 α : the child protection parameter

A higher α means a larger universal share of each newborn's grant. As $\alpha \rightarrow 1$, the system approaches a nearly pure per-child entitlement. As $\alpha \rightarrow 0$, the system becomes increasingly family-sensitive. Normatively, α governs how strongly society treats the child as an independent claimant.

20.2 μ : the family-size sensitivity parameter

A higher μ increases the rate at which the family-sensitive component declines with household size. It is a policy lever for budget control, but should not be raised so high that later-born children are effectively deprived.

20.3 K^* and λ : household taper design

K^* determines how many children receive full household weighting; λ controls the pace of taper afterward. Increasing K^* makes the family layer more generous to larger households. Increasing λ slows the taper.

20.4 ϕ and γ : the intergenerational tradeoff

A larger ϕ saves more current inflow for future generations but leaves less for present distribution. A larger γ increases current payouts from the fund but may weaken future capacity. These two parameters together encode a central intergenerational choice.

20.5 u_t : the moral floor

u_t is the non-negotiable child-development threshold. Reducing it to increase capital grants would strengthen formal asset ownership at the expense of present capability. Philosophically, that is usually the wrong trade unless the baseline service level is already high.

21 What the Framework Is and Is Not

It is useful to clarify what the SSCF does *not* claim.

1. It is not a claim that all inequality is unjust.
2. It is not a demand to liquidate all private wealth.
3. It is not a denial of family attachment, inheritance motives, or market activity.
4. It is not identical to universal basic income, though it can coexist with it.
5. It is not merely a baby-bonds proposal, because it integrates services, household support, fund design, demographic stabilization, and staged capability-linked access.

What it *is* is a pre-distributional institution: a way of ensuring that every person begins with a real material foothold in a world whose productive base is socially and historically inherited.

22 Governance, Legal Form, and Implementation Considerations

The mathematics alone is insufficient. The framework requires legal and institutional design.

22.1 Fund governance

The social wealth fund should be:

- legally ring-fenced,
- professionally managed,
- audited,
- protected from short-run political raids,
- transparent in asset allocation and performance.

The logic resembles intergenerational savings institutions already visible in Norway and, in partial citizen-distribution form, Alaska (Government of Norway [2025](#); State of Alaska, Permanent Fund Dividend Division [2026](#)).

22.2 Legal protection of child accounts

Child accounts should be:

- non-seizable by ordinary household creditors,
- protected from predatory guardianship,
- inheritable only under limited rules before adulthood,
- administered with a public fiduciary duty.

22.3 Administrative definitions

The model must define:

- who counts as an eligible child,
- how family/household status is determined,

- treatment of adoption, guardianship, and family dissolution,
- migration and citizenship/residency conditions,
- inflation indexation and regional cost multipliers.

These are implementation problems, not objections to the core architecture.

23 Limitations and Future Extensions

The framework remains a theoretical proposal. Several extensions are needed.

First, the macroeconomic effects of a large-scale social capital program require calibration: labor supply, fertility effects, educational attainment, entrepreneurship, asset prices, and housing markets would all matter.

Second, more elaborate demographic rules may be needed where population dynamics are volatile or where eligibility changes through migration.

Third, the top-up function could incorporate disability, regional cost, or care burdens. In the present paper these are suppressed for clarity.

Fourth, the relation between private inheritance and social birthright deserves deeper treatment. The framework is compatible with private bequests, but it may require stronger taxation of very large transfers if the goal is to reduce dynastic concentration.

Fifth, the education layer invites empirical research. Which modules matter? At what ages? How should paternalism be avoided while protecting young adults from fraud and coercion?

Finally, political economy matters. Any system capable of reducing structural inequality will face resistance from those who benefit from inherited concentration. A mathematically coherent framework is necessary, but never sufficient on its own.

24 Conclusion

This paper has argued for a simple but far-reaching claim: if the modern world is not built from scratch by each generation, then some part of its wealth cannot be treated as morally exhausted by current private title. The earth, institutions, infrastructure, legal order, public knowledge, and productive system are cumulative inheritances. To deny every new person a material share in that inheritance is to defend not justice, but late-entry exclusion.

The philosophical argument began with the birth lottery and traced its significance through competing theories of property and justice. Locke helps explain why labor

matters; Rousseau explains why property generates civil inequality; Paine explains why private appropriation of a common earth implies compensation; George explains why land rent is special; Marx and Polanyi explain why markets rest on historical restructuring; Rawls explains why morally arbitrary birth conditions require institutional correction; Sen and Van Parijs explain why substantive freedom requires material means; Ostrom explains why common wealth can be governed institutionally; Atkinson and Hamilton explain why capital endowments remain a live and serious policy idea.

The mathematical framework then translated those ideas into a dynamic institutional design. The SSCF combines: a universal child floor, a household allocation rule with taper, protected birthright accounts, progressive optional top-ups, education-linked staged access, and a sovereign social wealth fund. Its core ethical achievement is that every child retains a non-zero claim. Its core fiscal achievement is that family-related support is bounded and the fund can be managed under explicit draw constraints.

What the framework offers, then, is not utopia. It offers a serious alternative to the moral fiction that equal opportunity can exist without equal starting stakes. If a society truly believes that human worth is not determined by birth, then it must build institutions that make that claim materially real.

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